or expressed sequences contain biological informations in the matured processed mRNA. **Introns** are intervening sequences, which are non-coding sequences (non-amino acid-coding sequences) that should be removed from a gene before the mRNA product is made. Introns do not code for any enzyme or structural protein or polypeptides. These exons and introns are known as Split Genes.

3.8.2 RNA Splicing in plants

RNA Splicing is a process which involves the cutting or removing out of introns and knitting of exons. This process takes place in spherical particles which is a multiprotein complex called **SPLICISOMES.** It is approximately 40 – 60 nm in diameter. The spliceosomes have many small nuclear ribonucleic acids (snRNAs) and small nuclear ribonuclear protein particles (snRNPs) which identify and helps in the removal of introns.

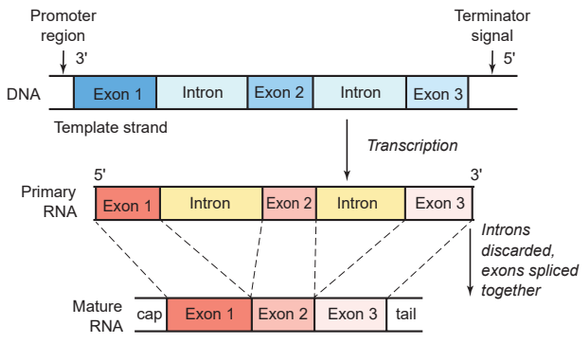


Figure: 3.32 RNA Splicing in plants

A spliceosome removes the introns with an enzyme ribozyme. Now the mature mRNA comes away from the spliceosomes through the nuclear pore and is transported out from the nucleus into cytoplasm, and gets attached to ribosome to carry out translation. The RNA and many proteins are transported through a nuclear pore complex by an energy dependent process.

3.8.3 Translation

The genetic information in the DNA code is copied onto mRNA bound in ribosomes for making polypeptides. The mRNA nucleotide sequence is decoded into amino acid sequence of the protein which is catalyzed by the ribosome. This process is called translation.

Terminology in Protein synthesis

Codon – DNA codes are referred to as triplet codes and those in mRNA is called as Codons. Each triplet specifies a particular amino acid. Codons present in mRNA are read in 5' → 3' direction. There are 64 codons of which 61 codons codes for amino acids.

Start codon – AUG specified methionine

Stop or Termination codon – UAA – Ochre UAG – amber and UGA – Opal.

Anticodons – The triplet of bases in a tRNA molecule is known as anticodon. In tRNA sequence of three bases which is complementary to codons of mRNA are called anticodon. The codons of mRNA are recognized by the anticodons of tRNA which are oriented in 3' → 5' direction.

Process of translation

The following are major steps in translation process

1. Initiation

The translation begins with the AUG codon (start codon) of mRNA. Translation occurs on the surface of the macromolecular arena called the ribosome. It is a nonmembranous organelle. During the process of translation the two subunits of ribosomes unite (combine) together and hold mRNA between them. The protein synthesis begins with the reading of codons of mRNA. The tRNA brings amino acid to the ribosome, a molecular machine which unites amino acids into a chain according to the information given by mRNA. rRNA plays the structural and catalytic role during translation.

A ribosome has one binding site for mRNA and two for tRNA. The two binding sites of tRNA are

i. **P-Site – The peptidyl – tRNA binding site** is one of the tRNA binding site. At this site

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